

Nguyen Mai Ngoc Hai

Intern Unity Developer

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[GitHub](#) | [Itch.io](#) | [Youtube](#) | [Portfolio](#)

PROFILE

Dedicated Gameplay Programmer with a strong foundation in C# and OOP principles, specializing in Unity 2D development. Proven expertise in leveraging robust design patterns (**Singleton**, **State**, **Object Pooling**) to architect scalable, high-performance game systems.

EDUCATION

FPT University

Bachelor of Game Programming

2022 - Present

WORK EXPERIENCE

Digital Transformation Laboratory (DX-Lab)

FPT University

Unity Developer

01/2026 - 04/2026

- Cross-Functional Integration:** Implemented core mechanics from concept to release in close collaboration with UI/UX and Design teams.
- Workflow Management:** Enforced Agile practices and strict Git/GitHub standards to optimize team collaboration and code integration.
- Gameplay Engineering:** Architected scalable game logic, AI, and UI systems in C# utilizing advanced design patterns (Singleton, State).

FEATURED PROJECTS

Brainrot Runner | Team Project | Unity Developer

Endless-runner featuring procedural gates and boss evolution cycles

- Gameplay Mechanics:** Implemented core runner mechanics and procedural gate generation utilizing Object Pooling to optimize performance.
- AI & Systems:** Programmed dynamic boss evolution cycles and enemy behaviors applying the State Pattern in C#.
- UI Integration:** Developed responsive in-game UI and Canvas systems to ensure seamless player feedback and state transitions.

Behind the Mask | Global Game Jam 2026 | Team Project

International 36-hour challenge delivered under extreme time constraints

- **Concept & Design:** Co-designed core gameplay loops and narrative concepts under extreme time constraints.
- **Asset Production:** Developed and integrated game models and visual assets, optimizing them for rapid prototype deployment.
- **Gameplay Implementation:** Programmed core logic and mechanics in C# to ensure a fully playable build within the 36-hour window.

Heroic Defense | Personal Project

2D Tower Defense

- **Core Mechanics:** Programmed robust tower placement and dynamic upgrade systems in C# to drive gameplay progression.
- **Systems Engineering:** Engineered internal resource management logic and wave-based enemy spawning systems.
- **UI Integration:** Developed intuitive UI/Canvas elements for real-time resource tracking and player interactions.

JUNGLE SURVIVOR | Personal Project | Demo

2D Action Platformer

- **Core Gameplay Loop:** Engineered the core survival mechanics and dynamic character progression systems in C#.
- **Combat & Physics:** Implemented fluid 2D physics and responsive combat mechanics to ensure a smooth, impactful game feel.
- **Systems Engineering:** Architected scalable logic for randomized upgrade selections and enemy behaviors to optimize the gameplay loop.

SKILLS & KNOWLEDGES

- **Programming & Architecture**
 - Architecting scalable systems applying **OOP**, **SOLID** principles, and advanced design patterns in C#.
- **Gameplay & Unity Engine**
 - Implementing core logic, dynamic AI behaviors, and 2D physics utilizing Unity.
- **UI & Database Management**
 - Developing responsive interfaces and structuring game data via **Canvas**, **SQL**, and **MySQL**.
- **Asset Optimization**
 - Editing 2D sprites and streamlining visual asset pipelines utilizing **Photoshop** and **AI Generation Tools**.
- **Workflow & Integration**
 - Streamlining team collaboration through Agile practices and version control (**Git/GitHub**).

CERTIFICATE

- **Game Development with Unity**
Codegym